

Two at a time

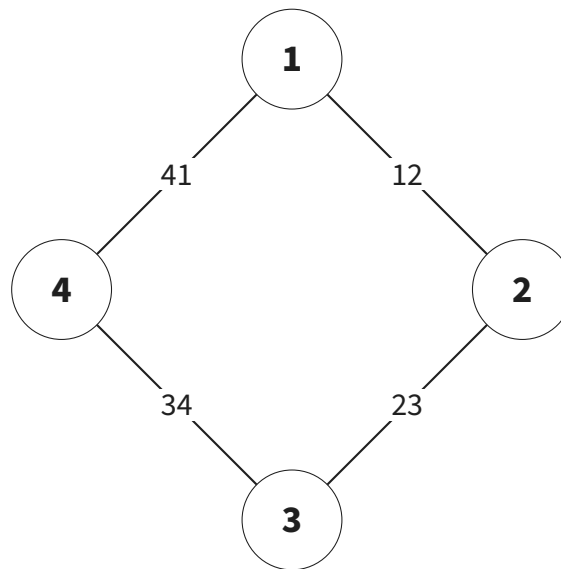
Name: _____

Date: _____

Put a two-sided counter on each lamp. White means OFF; dark means ON. One move: choose a line and **flip both counters at its ends**. Keep the counters in their places.

1. Learn the switches.

Start all OFF. Try a line. Try that same line again. What happened? Try any three moves and then undo them.



2. Make these lights.

Start all OFF each time. Can you make just the top and right lamps ON? What about just the top and bottom?

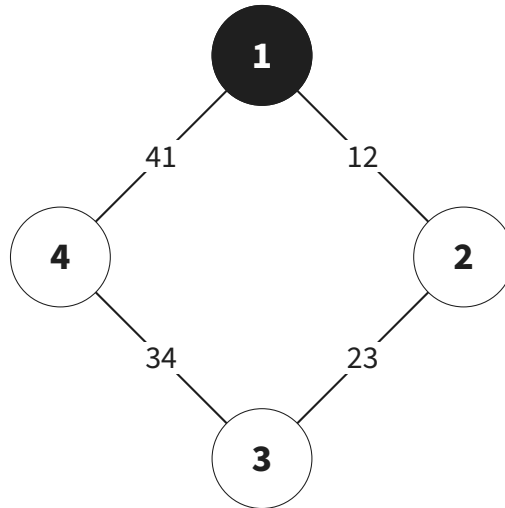
Adult: draw or write a move the child wants to remember.

Try it with objects. Explain by pointing, talking, or drawing.

A lamp that will not listen

3. Can you light just one?

Start all OFF. Try to leave only the top lamp ON. Try a few ideas, then pause and look for a reason.



A useful check: after each move, pair up the ON lamps. Is one always left over, or can they always make pairs?

4. A switch with a new rule.

Ask an adult to add a button that flips **just one** lamp. Now can you make the picture above? Which rule made the difference?

Show or tell what you noticed. An adult may record it.

Try it with objects. Explain by pointing, talking, or drawing.